

Q.1

(a) 7-Queen Problem and also write problem Formulation.

Problem Formulation

- 1) Initial state: An empty 7×7 chess board with no queens placed
- 2) State Space: All possible ways ^{to place} 0 to 7 queens on the board such that none attack each other
- 3) Actions: Place a queen in the next row in any column where it is not attacked by already placed queens
- 4) Transition Model: From state k queens, generate a state with $k+1$ queens placed legally.
- 5) Constraints
Queens can attack in horizontal, vertical and diagonal directions. So:
 - No ^{two} queens can be ~~on~~ in the same row.
 - No ^{two} queens can be in the same column
 - No two queens can be on the same diagonal
- 6) Goal state:
Exactly 7 queens are placed on the board with no pair of queens attacking each other
- 7) Path cost: Each placement has the same cost (can be ignored for this problem)

	0	1	2	3	4	5	6
0		Q ₁					
1				Q ₂			
2						Q ₃	
3	Q ₄						
4							Q ₅
5					Q ₆		
6			Q ₇				

Alpha - Beta pruning

$$V \geq 10$$

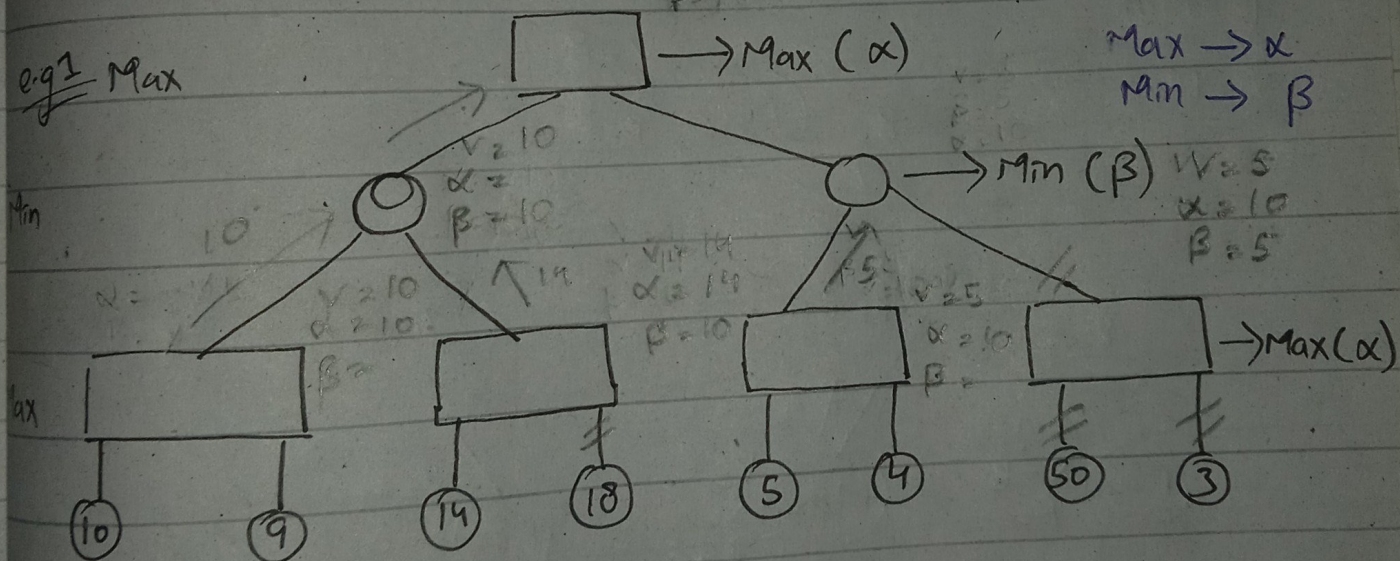
$$\alpha \geq 10$$

$$\beta \geq 1$$

Rule of pruning

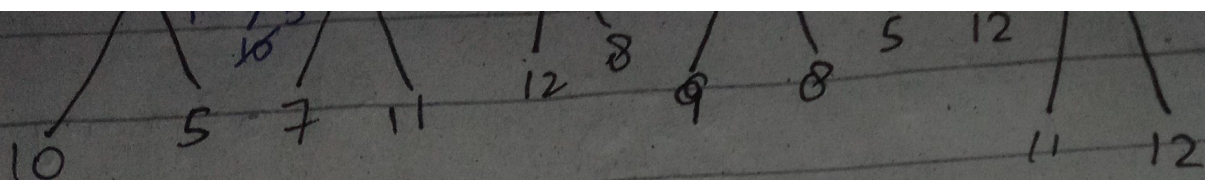
$$\alpha \geq \beta$$

e.g. 1 Max



$$10 \geq 5$$

Optimal path is 10



$\beta = \min$

$\alpha = \max$

$v = 8$

$\alpha = 8$

$\beta = 8$

8

$v = 8$

$\alpha = 8$

$\beta = 8$

8

$v = 8$

$\alpha = 8$

$\beta = 8$

9

$v = 9$

$\alpha = 9$

$\beta = 8$

2

$v = 2$

$\alpha = 8$

$\beta = 2$

$8 \geq 2$

2

$v = 2$

$\alpha = 8$

$\beta = 2$

Max

4

8

9

3

2

-2

9

-1

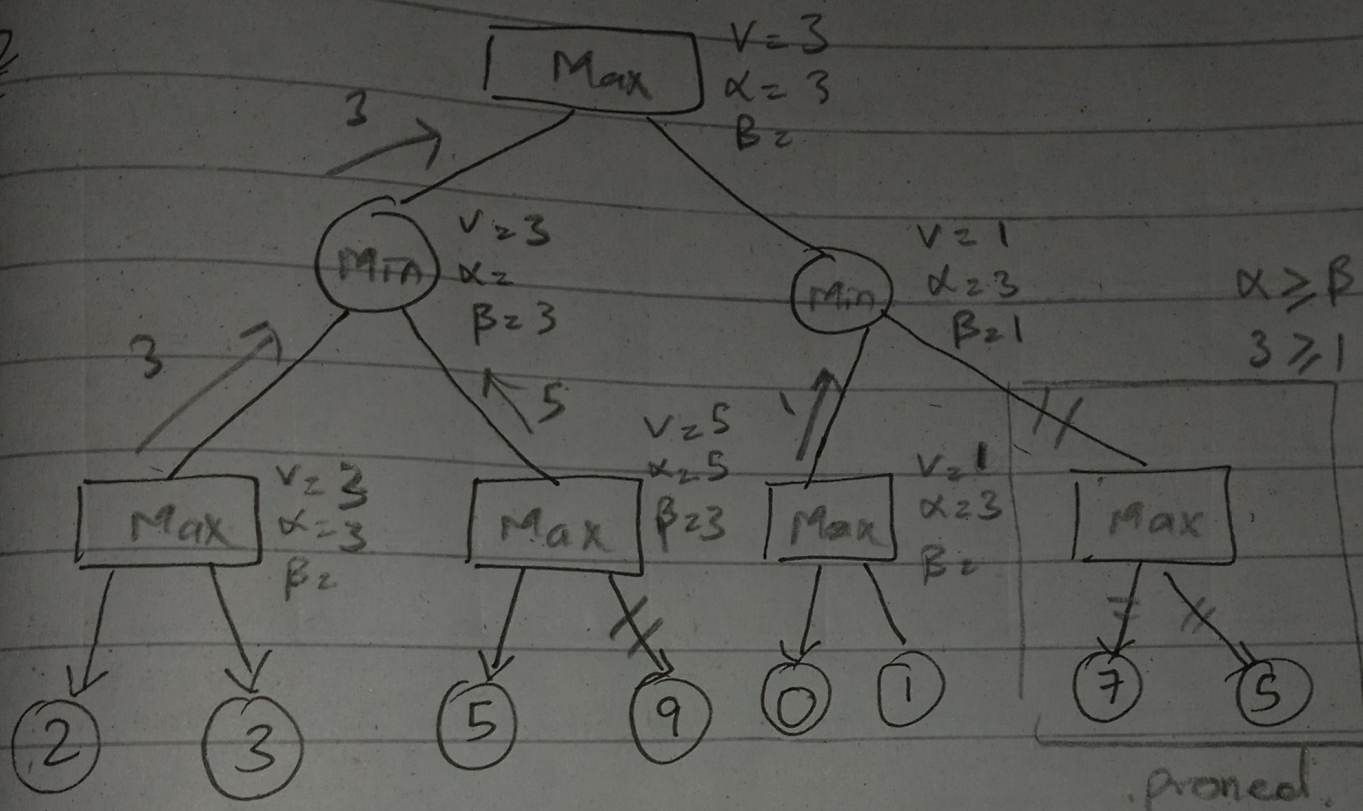
pruned

optimal path = 8

$\alpha \geq \beta$

$9 \geq 8$

Ex 2



$$\alpha \geq \beta$$

$$5 \geq 3$$

Optimal path = 3

Propositional logic - PL

10/15 marks

A propositional logic is a statement that is either true or false but not both

Connectives:

- ① $\wedge \rightarrow$ and $\rightarrow$ Conjunction
- ② $\vee \rightarrow$ or $\rightarrow$ Disjunction
- ③ $\neg \rightarrow$ Not $\rightarrow$ Negation ($\sim$)
- ④ $\Rightarrow$ If then $\rightarrow$ Implication ($\rightarrow$)
- ⑤ $\Leftrightarrow$ if and only if $\rightarrow$ Equivalence ($\Leftrightarrow$)

And

① $\wedge$ Conjunction

P	q	$P \wedge q$
T	T	T
T	F	F
F	T	F
F	F	F

P $\wedge$
 Its Monday and
 q
 Its raining
 $P \wedge q$

② OR ($\vee$) $\rightarrow$ Disjunction

P	q	$P \vee q$
T	T	T
T	F	T
F	T	T
F	F	F

P $\vee$
 Its Monday OR
 q
 Its Raining
 $P \vee q$

Not (Negation) $\neg$ ($\sim$)

P	$\neg P$
T	F
F	T

logic is easy $\rightarrow P$
 logic is not easy $\rightarrow \neg P$

Implication If then ($\rightarrow$)

P	Q	$P \rightarrow Q$
T	T	T
T	F	F
F	T	T
F	F	T

If I study hard then
 I get rich

$P \rightarrow Q$

Equivalence ($\Leftrightarrow$)

P	Q	$P \Leftrightarrow Q$
T	T	T
T	F	F
F	T	F
F	F	T

If I have triangle, then
 my polygon has only
 three triangle

x)

Exercise:

Construct a truth table for

① $(A \vee B) \rightarrow C$

② $(P \vee q) \wedge \neg r$

③ $\neg((P \rightarrow q) \vee (q \rightarrow P))$

①	A	B	C	$A \vee B$	$(A \vee B) \rightarrow C$
	T	T	T	T	T
	T	T	F	T	F
	T	F	T	T	T
	T	F	F	T	F
	F	T	T	T	T
	F	T	F	T	F
	F	F	T	F	T
	F	F	F	F	T

②	P	q	r	$P \vee q$	$(P \vee q) \wedge \neg r$
	T	T	T	T	F
	T	T	F	T	T
	T	F	T	T	F
	T	F	F	T	T
	F	T	T	T	F
	F	T	F	T	T
	F	F	T	F	F
	F	F	F	F	F

③

P	q	$P \rightarrow q$	$q \rightarrow P$	$(P \rightarrow q) \vee (q \rightarrow P)$	$\neg$
T	T	T	T	T	F
T	F	F	T	T	F
F	T	T	F	T	F
F	F	T	T	T	F

Q) My breakfast is either eggs or cereal and toast.

P $\vee$ q

R and R

$P \vee (q \wedge R)$

P	q	R	$q \wedge R$	$P \vee (q \wedge R)$
T	T	T	T	T
T	T	F	F	T
T	F	T	F	T
T	F	F	F	T
F	T	T	T	T
F	T	F	F	F
F	F	T	F	F
F	F	F	F	F

Propositional logic

Tautology $\rightarrow$ is a proposition which is always True

Contradiction $\rightarrow$ is a proposition which is always False

contingency $\rightarrow$ is a proposition which is either a True and a False

① $P \rightarrow (P \vee q) \rightarrow$ prove that $2^2 = 4$

P	q	$P \vee q$	$P \rightarrow (P \vee q)$
T	T	T	T
T	F	T	T
F	T	T	T
F	F	F	T

Tautology
proved

② $(P \vee q) \wedge (\neg P \wedge \neg q)$

P	q	$P \vee q$	$\neg P \wedge \neg q$	$(P \vee q) \wedge (\neg P \wedge \neg q)$
T	T	T	F	F
T	F	T	F	F
F	T	T	F	F
F	F	F	T	F

Contradiction
proved

First Order Logic (FOL)

FOL is a way of writing and understanding statements so that we can describe things clearly and logically. It helps us talk about objects, their properties and their relationships with each other.

FOL (like natural language).

Objects $\Rightarrow$ People, house, numbers, color etc

Relations $\Rightarrow$ Red, round, prime

Variables $\rightarrow$ like (x , apple)

symbols $\rightarrow$ x, y, z etc

connectives $\rightarrow$ not ($\neg$), And ($\wedge$), OR ($\vee$), IF ($\rightarrow$),
Equivalence ($\leftrightarrow$).

Pre dictes $\Rightarrow$ Greater (1, 2), color of (basket)

$\hookrightarrow$ part of a sentence containing a work, action or something about the attribute

Quant.ifiers-

$\hookrightarrow$ $\forall$ $\exists$

Universal ($\forall x$) $\rightarrow$

Existential ($\exists x$) $\wedge$

It can handle more complex ^{conditions} ~~conditions~~ elements.

Q) Translating English Sentences into FOL :-

① All boys like cricket.

$\forall x :- \text{Boys}(x) \rightarrow \text{like}(\underbrace{x}_{\text{sub}}, \underbrace{\text{cricket}}_{\text{object}})$

② Some boys like cricket

$\exists x :- \text{Boys}(x) \wedge \text{like}(x, \text{cricket})$

③ Ali is tall

$\text{tall}(\text{Ali})$

④ Clinton is not tall

$\neg \text{tall}(\text{Clinton})$

⑤ Every Gardener likes the sun

$\forall x :- \text{Gardener}(x) \rightarrow \text{like}(x, \text{sun})$

⑥ Not all students like both Math and Science

$\neg \forall x :- \text{Students}(x) \rightarrow \text{like}(x, \text{Math}) \wedge \text{like}(x, \text{Science})$
 ~~$\text{like}(x, \text{math and science})$~~

⑦ All students like football

$x = \text{students}$ $(x) \rightarrow \text{like}(x, \text{football})$

Machine learning Algorithm (ML)

ML is a part of AI. It means teaching computers to learn from data and make decisions or predictions without directly programmed.

For example:

When you watch videos on youtube, it learns what you like and shows similar videos that ML in action.

Why ML is used in AI

- ① learn from past experience (Data)
- ② Improve Automatically over the time
- ③ Make smart decisions on their own.

Types of ML

- ① supervised learning
- ② unsupervised learning

~~KNN~~ • KNN Algorithm

• Naive Bayes Algorithm

Apriori Algorithm

unsupervised

Apriori Algorithm

The Apriori Algorithm used in AI to find patterns in large sets of data. It helps computer understand what items often go together. This is useful for making decisions, predictions or giving recommendations.

- ① Looking at a lot of data
- ② Finding items that appear together

Real life Examples

- ① Shopping Basket
- ② Marketing
- ③ Stock Management

Example ①

Suppose we ~~AA~~ have the following dataset that has various transactions, and from this dataset we need to find the frequent item sets and generate association rules using Apriori Algorithm. -

TID	Item sets	T ₆	B, C
T ₁	A, B	T ₇	A, C
T ₂	B, D	T ₈	A, B, C, E
T ₃	B, C	T ₉	A, B, C
T ₄	A, B, D		
T ₅	A, C		

Min Support = 2

Min confidence = 50%

Sol

Step ①

Item set	Support count
A	6
B	7
C	6
D	2
X E	1

↳ b/c less than min support

Step ②

Item set	Support count
A	6
B	7
C	6
D	2

Step ③

Item set	Support count		
A, B	4	C, D	0
A, C	4	X	
X A, D	1	↳ less than	
B, C	4	min support	
B, D	2		

Step 4

Itemset	Support count
A, B	4
A, C	4
B, C	4
B, D	2

Step 5

Item set	Support count
A, B, C	2
X A, B, D	1
X A, C, D	0
X B, C, D	0

Step 6

Item set	Support count
A, B, C	2

Step 7 Min ~~support~~ confidence = 50%

Association Rule: 

Rules	Support	confidence
$A \cap B \rightarrow C$	2	$2/4 = 0.5 = 50\%$
$B \cap C \rightarrow A$	2	$2/4 = 0.5 = 50\%$
$A \cap C \rightarrow B$	2	$2/4 = 0.5 = 50\%$
$C \rightarrow A \cap B$	2	$2/6 = 0.33 = 33\%$ X
$A \rightarrow B \cap C$	2	$2/6 = 0.33 = 33\%$ X
$B \rightarrow A \cap C$	2	$2/7 = 0.28 = 28\%$ X less than 50%

$A \cap B$
 $A \cap C$
 $B \cap$

short

Fuzzy
thing
a S
logi
con

For

In

In

M

1) An e
not

1)

$$A \wedge B \rightarrow C$$

$$A \wedge C \rightarrow B$$

$$B \wedge C \rightarrow A$$

Strong Association

Fuzzy logic

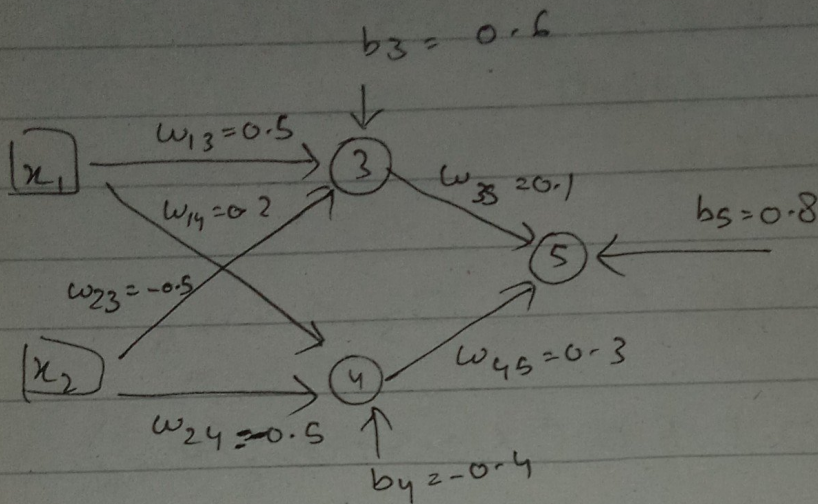
Fuzzy logic is a way of thinking that allows things to be partly True and partly False at a same time. It is different from Normal logic ^{called} ~~and~~ binary logic) where things either completely True and False

For example

In Normal logic: A person is either tall or not tall.

In Fuzzy logic: A person can be a little tall, medium tall, very tall (Degree of truth)

Back Propagation Numerical



①)

Input (x_1)	x_2	Output b
1	1	0

Learning rate (ϵ) = 0.5

Sigmoid Function $\sigma = \frac{1}{1 + e^{-x}}$

↓

Output

Step 01) For hidden layer 1 Input

$$N_3 = x_1 \times w_{13} + x_2 \times w_{23} + b_3$$

$$N_3 = 1 \times 0.5 + 1 \times -0.5 + 0.6$$

$$N_3 = 0.2$$

$$\text{Output} = \frac{1}{1 + e^{-x}} = \frac{1}{1 + e^{-0.2}}$$

$$= 0.54$$

N_4

$$N_4 = x_1 \times w_{14} + x_2 \times w_{24} \times b_4$$

$$= 1 \times 0.2 + 1 \times (-0.5) \times (-0.4)$$
$$= 0.4$$

$$\text{output} = \frac{1}{1+e^{-x}} = \frac{1}{1+e^{-0.4}} = 0.59$$

$$N_5 = O_3 \times w_{35} + O_4 \times w_{45} \times b_5$$

$$= 0.54 \times 0.1 + 0.59 \times 0.3 \times 0.8$$
$$= 0.19$$

$$\text{output} = \frac{1}{1+e^{-x}} = \frac{1}{1+e^{-0.19}} = 0.54$$

Step 2 calculation of error

$$\text{Formula: } \text{Err}_j = O_j (1 - O_j) (6 - O_j)$$

For output layer

$$\text{Err}_5 = O_5 (1 - O_5) (6 - O_5)$$

$$= 0.54 (1 - 0.54) (0 - 0.54)$$
$$= -0.13$$

For hidden layer

$$\begin{aligned}\text{Err}_4 &= O_4 (1 - O_4) (0 - O_4) \\ &= 0.59 (1 - 0.59) (0 - 0.59)\end{aligned}$$

$$\text{Err}_4 = -0.14$$

$$\begin{aligned}\text{Err}_3 &= O_3 (1 - O_3) (0 - O_3) \\ &= 0.54 (1 - 0.54) (0 - 0.54) \\ &= -0.13\end{aligned}$$

Step 3 ~~we~~ update the weights

Formula

$$W_j(\text{new}) = W_j(\text{old}) + I \times O_i \times \text{Err}_j$$

$$\begin{matrix} w_{45} \\ w_{35} \end{matrix} = \left[\begin{array}{c} \text{output layer} \\ O_4 \\ O_3 \end{array} \right] \text{Err}_5$$

$$\begin{matrix} w_{24} \\ w_{14} \end{matrix} = \left[\begin{array}{c} \text{hidden} \\ \text{layer} \\ O_2 \\ O_1 \end{array} \right] \text{Err}_4$$

$$\begin{matrix} w_{23} \\ w_{13} \end{matrix} = \left[\begin{array}{c} \text{input} \\ \text{layer} \\ O_2 \\ O_1 \end{array} \right] \text{Err}_3$$

$$\begin{aligned}
 w_{45}^{(new)} &= w_{45}^{(old)} + \tau \times O_4 \times \text{Err}_5 \\
 &= 0.3 + 0.5 \times 0.59 \times (-0.13) \\
 &= 0.26
 \end{aligned}$$

$$\begin{aligned}
 w_{35}^{(new)} &= w_{35} + \tau \times O_3 \times \text{Err}_5 \\
 &= 0.1 + 0.5 \times 0.54 \times (-0.13) \\
 &= 0.06
 \end{aligned}$$

$$\begin{aligned}
 w_{24} &= w_{24} + \tau \times O_2 \times \text{Err}_4 \\
 &= (-0.5) + 0.5 \times 1 \times (-0.14) \\
 &= -0.57
 \end{aligned}$$

$$\begin{aligned}
 w_{14} &= w_{14} + \tau \times O_2 \times \text{Err}_4 \\
 &= 0.2 + 0.5 \times 1 \times (-0.14) \\
 &= 0.13
 \end{aligned}$$

$$\begin{aligned}
 w_{23} &= w_{23} + \tau \times O_2 \times \text{Err}_3 \\
 &= (-0.3) + 0.5 \times 1 \times (-0.13) \\
 &= -0.36
 \end{aligned}$$

$$\begin{aligned}
 w_{13} &= w_{13} + \tau \times O_1 \times \text{Err}_3 \\
 &= 0.5 + 0.5 \times 1 \times (-0.13) \\
 w_{13} &= 0.43
 \end{aligned}$$

$$b_5 = \text{Err}_5$$

$$b_4 = \text{Err}_4$$

$$b_3 = \text{Err}_3$$

Step 04 update the bias value

Formula :

$$b_j^{(\text{new})} = b_j^{(\text{old})} + I \times \text{Err}_j$$

$$b_{5\text{-new}} = b_5 + I \times \text{Err}_5$$

$$= 0.8 + 0.5 \times (-0.13)$$

$$= 0.73$$

$$b_4 = b_4 + I \times \text{Err}_4$$

$$= -0.4 + 0.5 \times (-0.14)$$

$$= -0.5$$

$$b_3 = b_3 + I \times \text{Err}_3$$

$$= 0.1 + 0.5 \times (-0.13)$$

$$= 0.54$$